
Input:

q — Exponent of the Lebesgue space of interest
 $\mathbf{u}^{\text{init}}$ — Initial guess corresponding to the small-data limit (see Section 3)
 B^{init} — Initial value for the constraint parameter
 B^{max} — Maximum value for the constraint parameter
 δB — Increment of the constraint parameter
 ϵ — Convergence tolerance

Output:

Extreme states $\tilde{\mathbf{u}}^B$ and corresponding values of the objective functional $\mathcal{R}(\tilde{\mathbf{u}}^B)$ for $B^{\text{init}} \leq B \leq B^{\text{max}}$

Function MainBranch():

```
 $B \leftarrow B^{\text{init}}$   
 $\mathbf{u}_0 \leftarrow \mathbf{u}^{\text{init}}$   
repeat  
   $\tilde{\mathbf{u}}^B \leftarrow \text{SolveProblem3}(q, B, \mathbf{u}_0, \epsilon)$   
  Evaluate  $\mathcal{R}(\tilde{\mathbf{u}}^B)$  according to (15)  
   $\mathbf{u}_0 \leftarrow \tilde{\mathbf{u}}^B$  // Branch continuation with prev max  
   $B \leftarrow B(1 + \delta B)$  // Increase constraint value  
until  $B \geq B^{\text{max}}$ 
```

Function SolveProblem3($q, B, \mathbf{u}_0, \epsilon$):

```
 $k = 0$   
 $\Gamma_{\tau_{-1}\mathbf{d}_{-1}}(\mathbf{d}_{-1}) = \mathbf{0}$   
repeat  
  Compute the  $L^2$  gradient  $\nabla^L \mathcal{R}(\mathbf{u}_k)$  according to (36)  
  Compute the  $H^{3/2-1/q}$  gradient  $\nabla^H \mathcal{R}(\mathbf{u}_k)$  according to (37)  
  Compute the step direction  $\mathbf{d}_k$  according to (44)  
  Find step size  $\tau_k$  by solving (40)  
  Compute  $\mathbf{u}_{k+1}$  by (43)  
  Compute the vector transport  $\Gamma_{\tau_k \mathbf{d}_k}(\mathbf{d}_k)$  by (41) for the next  
  iteration  
   $k \leftarrow k + 1$   
until  $|\mathcal{R}(\mathbf{u}_{k+1}) - \mathcal{R}(\mathbf{u}_k)| / |\mathcal{R}(\mathbf{u}_k)| < \epsilon$   
return  $\mathbf{u}_{k+1}$ 
```

Algorithm 1: Continuation approach used to compute branches of local maximizers $\tilde{\mathbf{u}}^B$ for increasing values of the constraint parameter B .